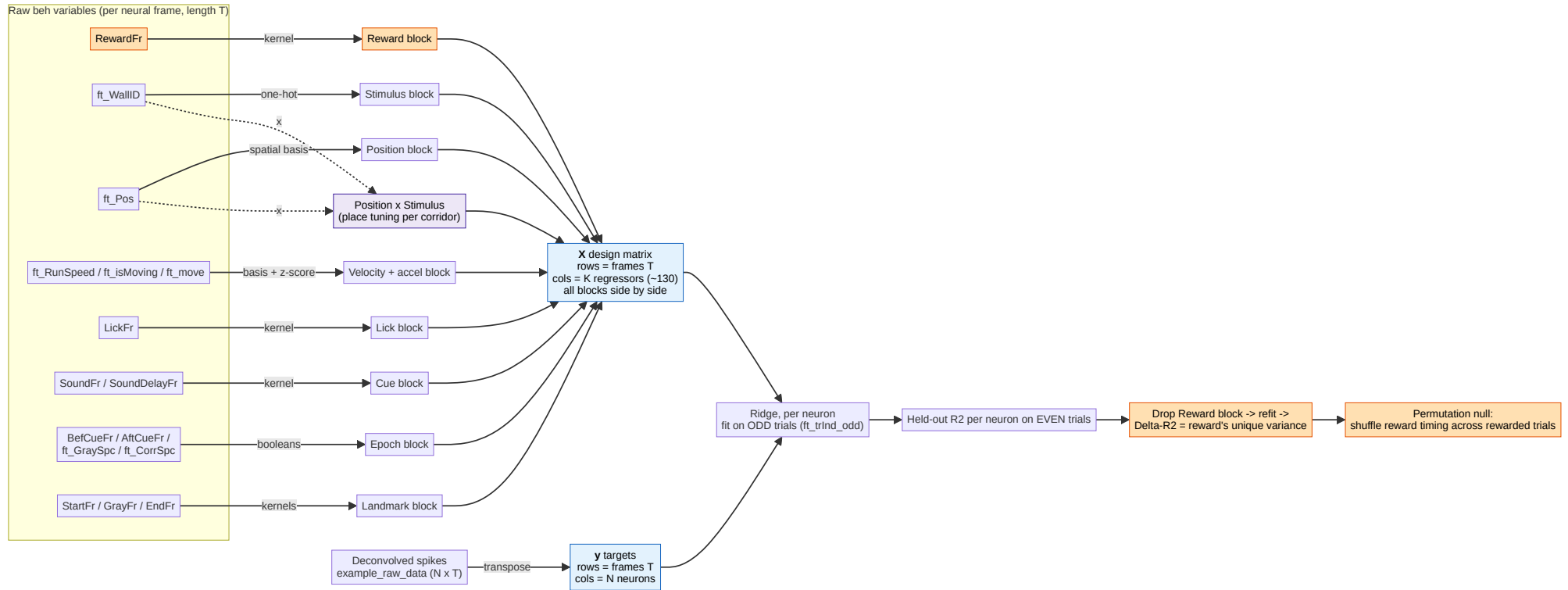


Encoding model: building the (augmented) design matrix X and targets y

Zhong et al. 2025 — per-neuron ridge encoding model. Many regressors (K large) is fine: there are far more frames T than features, and ridge controls variance. X ($T \times K$) and y ($T \times N$) share the same frame rows T.



Augmented feature catalog — every regressor block in X

Each raw beh variable is expanded into a block of number-columns. Only frame-aligned (ft_... / ...Fr) variables are used, so every block lines up row-for-row with the neural frames. Column counts are typical; exact counts depend on the session (number of unique corridors, chosen number of basis bins/kernel lags).

Block	beh source	Encoding (how it becomes numbers)	~cols	Why
Stimulus identity	ft_WallID	one-hot over every corridor in UniqWalls (circle1/2, leaf1/2/3, swaps)	5–7	which texture is showing
Position	ft_Pos	12 spatial basis bins across 0–60 (raised-cosine or one-hot)	12	where in the corridor
Position × Stimulus	ft_Pos × ft_WallID	position bins repeated per corridor	~60	place tuning per corridor — the big, biologically-motivated expansion
Running speed	ft_RunSpeed / RunFr	z-scored value + 5 raised-cosine basis functions	6	locomotion strongly drives visual cortex
Movement state	ft_isMoving	boolean	1	moving vs still
Acceleration	diff(ft_RunSpeed) / ft_move	z-scored continuous	1	speed changes
Reward ← of interest	RewardFr	8-lag raised-cosine kernel around delivery frame	8	reward response ramps & fades
Lick	LickFr	8-lag kernel	8	motor/consummatory control (critical to separate from reward)
Sound cue	SoundFr / SoundDelayFr	8-lag kernel	8	cue-evoked response (predicts reward)
Cue epoch	BefCueFr / AftCueFr	2 booleans	2	sustained pre- vs post-cue state
Space epoch	ft_GraySpc / ft_CorrSpc	2 booleans	2	textured corridor vs gray space
Corridor landmarks	StartFr / GrayFr / EndFr	6-lag kernel each	~18	entry / gray-onset / exit transients
Total design matrix width K			~130	vs T = tens of thousands of frames → well-conditioned with ridge

Kept from before: target y = deconvolved spikes; split train/test by trial (ft_trInd_odd / ft_trInd_even), never by frame; ridge with λ tuned on training folds; reward importance = drop the reward block and measure held-out ΔR^2 .

Worked example: 10 frames of the augmented X (representative columns)

Two trials — a leaf1 (rewarded) corridor then a circle1 (never-rewarded) corridor. Each row is one imaging frame. "..." marks the other columns of that block (not shown for space).

Blocks: stimulus position pos×stim velocity reward lick cue epoch

frame	S_leaf1	S_circ1	...	P_mid	...	leaf1×P_mid	...	Vel	Accel	Moving	Rew_0	Rew_1	...	Lick_0	...	Cue_0	AftCue	Gray
0	1	0	...	0	...	0	...	5	0	1	0	0	...	0	...	0	0	0
1	1	0	...	0	...	0	...	6	1	1	0	0	...	0	...	1	1	0
2	1	0	...	1	...	1	...	6	0	1	0	0	...	0	...	0	1	0
3	1	0	...	1	...	1	...	4	-2	1	1	0	...	1	...	0	1	0
4	1	0	...	0	...	0	...	3	-1	1	0	1	...	0	...	0	1	0
5	1	0	...	0	...	0	...	2	-1	1	0	0	...	0	...	0	1	1
6	0	1	...	0	...	0	...	5	0	1	0	0	...	0	...	0	0	0
7	0	1	...	0	...	0	...	6	1	1	0	0	...	0	...	1	1	0
8	0	1	...	1	...	0	...	5	-1	1	0	0	...	1	...	0	1	0
9	0	1	...	0	...	0	...	3	-2	1	0	0	...	0	...	0	1	1

y — targets (same 10 frames, one column per neuron)

frame	neuron 1	neuron 2	...
0	0.2	0.5	...
1	0.3	0.6	...
2	0.4	0.6	...
3	0.9	0.4	...
4	0.7	0.3	...
5	0.2	0.2	...
6	0.2	0.5	...
7	0.2	0.6	...
8	0.2	0.5	...
9	0.1	0.3	...

Two things this widened table shows. (1) **Position × Stimulus interaction:** leaf1×P_mid is 1 only when the mouse is in the leaf1 corridor AND at the middle position (frames 2–3). At frame 8 the mouse is also at P_mid but in the *circle* corridor, so the leaf1 interaction column is 0 — this is how the model learns a separate place-field per corridor. (2) **Reward vs lick are decoupled:** at frame 3 both reward and lick fire together (rewarded corridor), but at frame 8 the mouse licks with **no reward** (wrong corridor). Because both blocks are in the model, the reward block only earns credit for the extra activity at real reward — that is what makes the ΔR^2 a clean reward signal rather than a lick signal.